

Machine Learning and Computer Simulation

Disentangle the Fuzzy Inhibitor Binding by Hsp90

Mohammad Sahil, Jaya Krishna Koneru, and Jagannath Mondal*

*Tata Institute of Fundamental Research Hyderabad, 36/P Gopanapalli Village, TS-500046,
India*

E-mail: jmondal@tifrh.res.in

Abstract

Heat shock protein 90 (Hsp90) is an essential molecular chaperone that plays a critical role in protein folding, stabilization, and activation, positioning it as a prominent target in cancer therapy. The activation of Hsp90 is stringently controlled by conformational dynamics within its N-terminal domain (N-Hsp90), which contains the ATP-binding site and serves as the main focus for the development of small-molecule inhibitors. However, despite the development of numerous small-molecule inhibitors, effectively targeting N-Hsp90 remains challenging and this difficulty arises from conflicting crystallographic and NMR structures. Here, through statistical analysis of aggregated 60 microseconds of all-atom unbiased Molecular Dynamics (MD) simulations, that capture various stages of the recognition process of the prominent inhibitor Geldanamycin (GDM) with N-Hsp90, we reveal that the inhibitor binding event is dictated by intrinsic conformational heterogeneity of loop-4 and especially its dynamic oscillation between loop-in and loop-out states. The complexity of this process arises from the conformational exploration of existing states and the generation of new loop-out states in both apo and inhibitor-bound N-Hsp90, suggesting the interplay of dual mechanisms. Using a machine learning-based approach, we quantitatively disentangle

the contributions of conformational-selection and induced-fit mechanisms. Our findings show that conformational-selection predominantly involves loop-4 and adjacent helices, while the induced-fit mechanism is localized to the binding pocket and loop-3, highlighting a spatially segregated dual binding pathway for inhibitor recognition. We anticipate that, by targeting specific transitions within the binding pathway, it may be possible to develop next-generation N-Hsp90 inhibitors with improved efficacy and selectivity.

Significance of statement

Hsp90's role as a molecular chaperone in cancers makes it a critical therapeutic target, yet the design of effective inhibitors remains a matter of ongoing research due to its intrinsic conformational heterogeneity and cytotoxicity of available inhibitors. This study integrates extensive molecular dynamics simulations with machine learning to deconstruct the complex binding pathway of Geldanamycin (GDM), uncovering a long-lived intermediate state that reconciles conflicting structural models. By quantitatively disentangling conformational-selection from induced-fit, we reveal a spatially segregated dual mechanism of inhibitor recognition, an insight with potential implications for drug discovery. By combining unbiased molecular simulations with adaptive sampling on a prototypical substrate, this work establishes a rigorous framework for investigating highly dynamic molecular chaperones.

Introduction

Heat shock protein 90 (Hsp90) is one of the most abundant cytosolic proteins in eukaryotes, accounting for 4-6% of the total cytosolic protein content under stress conditions.¹ As a ubiquitous molecular chaperone, Hsp90 plays a crucial role in facilitating the folding and stabilizing the higher-order assemblies of a wide range of client proteins.^{2,3} These clients rely on Hsp90 to maintain their functional conformations, especially under stress con-

ditions.⁴ Structurally, Hsp90 consists of three functional domains: the N-terminal domain (N-Hsp90), the middle domain, and the C-terminal domain.⁵ Hsp90 functions as a dimer, with the C-terminal domains forming a V-shaped dimeric structure that positions the client proteins between the two middle domains.³ The activation of Hsp90 is tightly regulated by conformational changes in the N-terminal domain, which houses an ATP-binding site. ATP binding induces conformational changes in both the N-terminal and middle domains, thereby promoting ATP hydrolysis and activating the chaperoning function of Hsp90.⁶

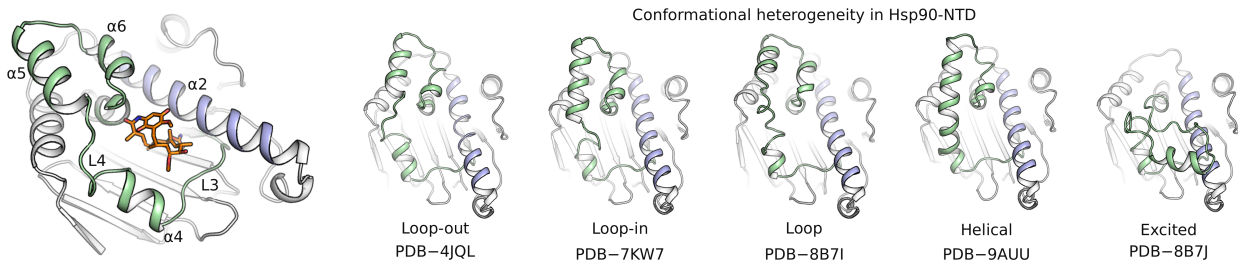


Figure 1: Conformational heterogeneity in Hsp90: The N-Hsp90 bound with one of the inhibitors (GDM) in its ATP binding site (PDB-1YET). Important secondary structures are noted, along with their different possible conformations observed crystallographically.

Hsp90’s chaperoning activity is essential not only for maintaining cellular homeostasis under normal and stressed conditions but also for supporting the survival of cancer cells, where elevated levels of Hsp90 are often observed.⁷ The uncontrolled growth and high stress levels in cancer cells necessitate the chaperoning function of Hsp90 to assist in the folding and stabilization of oncoproteins, while simultaneously inhibiting stress-induced apoptosis.¹ The critical role of Hsp90 in tumor cell survival has been established across various cancer types, including both solid tumors and hematological malignancies. Inhibiting or knocking out Hsp90 has been shown to suppress tumor growth, making N-Hsp90 a promising target for anti-cancer therapies.⁸

Given its potential in cancer treatment, numerous Hsp90 inhibitors have been developed and tested.¹ Geldanamycin (GDM), a natural compound extracted from *Streptomyces hygroscopicus*, was one of the first inhibitors to enter clinical trials.⁹ Although GDM effectively inhibited Hsp90, its clinical development was halted due to unacceptable toxicological

effects observed in vivo. However, several derivatives of GDM, such as 17-AAG and 17-DMAG, have since advanced to clinical trials. Another notable class of Hsp90 inhibitors includes Radicicol and its derivatives.¹ In addition to natural inhibitors, numerous synthetic inhibitors representing various chemical classes such as purines, pyrazoles, triazines, quinolines, coumarins, isoxazoles, and hybrids-are under investigation.¹⁰ To date, more than 100 inhibitors from these classes have been extracted or synthesized, but none have yet been finalized for clinical use.¹¹ Despite this, ongoing research continues to focus on the development of new derivatives, particularly for GDM, with over 40 derivatives currently known and more in development.¹²⁻¹⁵

Despite the significant interest in Hsp90 inhibitors, the molecular and atomistic basis of their binding mechanisms remains incompletely understood. Hundreds of crystal structures of N-Hsp90 bound to ATP or various inhibitors have been determined, consistently revealing a single binding site for substrate ATP or inhibitors (Fig. 1). While crystallography has provided valuable insights into the bound state, NMR studies of GDM suggest a two-step binding mechanism involving an intermediate state via an induced-fit mechanism.¹⁶ Conversely, a single-step binding mechanism has been proposed for the GDM derivative 17-DMAG,¹⁷ potentially indicating a conformational-selection route for inhibitor binding. As discussed below, the substrate-binding mechanism is further complicated by the strong conformational heterogeneity present in the Hsp90 binding pocket.

A key aspect of N-Hsp90 is the high conformational plasticity observed at loop-4 (L4 or the lid), which can broadly be classified into five conformations (Fig. 1): a closed conformation referred as loop-in whereby the L4 points towards binding pocket; similarly loop-out and loop conformations possessed L4 pointed outwards from binding pocket or in the middle; a more organized helical conformation whereby the L4 stabilizes as a continuous helix merging with $\alpha 4$ and $\alpha 5$; and a recently determined excited conformation whereby the entire L4, $\alpha 5$ and $\alpha 6$ significantly re-orient to cover the binding pocket completely.¹⁸⁻²¹ Further conformations exist for each class within the broader definitions provided here. Prior NMR²²

and MD²³⁻²⁵ studies have provided further evidence for the existence of multiple conformations of Hsp90 in its free form, prior to any binding event. Conformational plasticity is known to affect biomolecular recognition,²⁶ and it is crucial for inhibitor binding in Hsp90, with the entropic contribution of loop-4 playing a significant role.²⁷ Most Hsp90 inhibitors exhibit specificity for particular protein conformations, binding preferentially to one of the L4 states, and are thus referred to as ‘loop binders’, ‘helical binders’ etc. The substrate of interest in this study, the parent molecule GDM, specifically binds to the loop-out conformation, as observed in substrate-bound crystal structures (PDB ID: 1YET).²⁸ Despite extensive studies focusing on the end states of Hsp90 (i.e., apo or bound states),^{29,30} a detailed understanding of the binding process and its interplay with the conformational heterogeneity of Hsp90 is still lacking. For instance, whether GDM selects a pre-existing loop-out state (conformational-selection) or induces the loop-out state, or a combination of both, remains unclear.

In this work, we aim to elucidate the role of conformational heterogeneity in the binding of GDM to Hsp90 by simulating the recognition process of GDM within its native cavity. To this end, we conduct extensive unbiased molecular dynamics (MD) simulations, to track the search for the native binding pose of GDM in Hsp90, complemented by adaptively sampled trajectories which were seeded by extensive enhanced sampling of sparsely populated states. We employ a comprehensive Markov state modeling (MSM) approach to characterize the dynamic binding process of GDM to Hsp90, identifying a two-stage recognition mechanism facilitated by a binding intermediate. The mechanism was independently confirmed via other sampling routes. Additionally, a follow-up MSM analysis of both the apo and native substrate-bound states sheds light on the conformational heterogeneity near the L4 region, revisiting the debate on conformational-selection versus induced-fit. Finally, we resolve this debate by applying a machine learning based classifier^{31,32} to the simulated trajectories, quantifying the contributions of both mechanisms at play in the binding process.

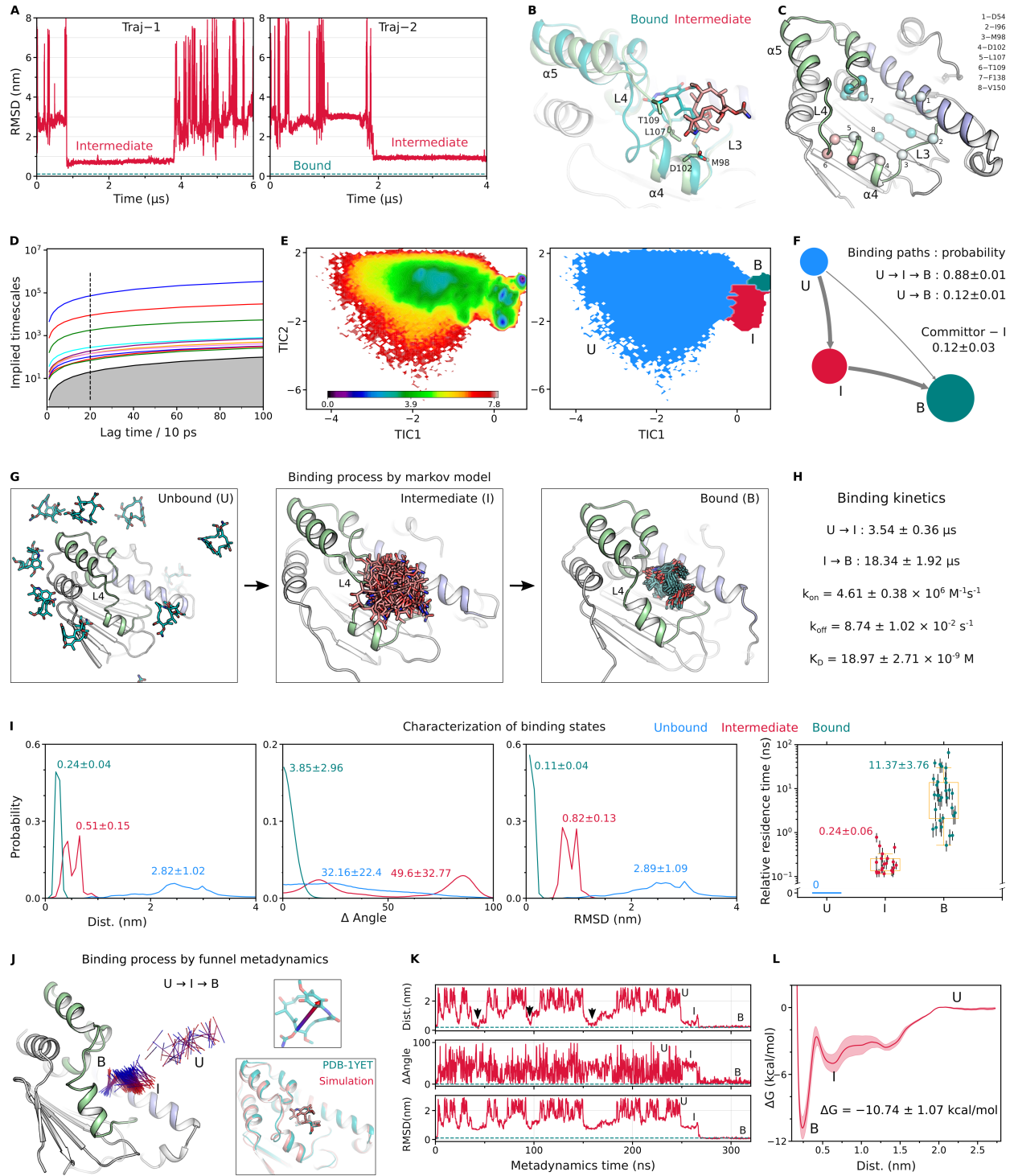


Figure 2: The inhibitor binding to N-Hsp90: (A) The RMSD timeprofile of GDM with respect to bound conformation as observed in unbiased binding simulations. The teal dashed line indicate RMSD values in bound simulations. (B) The simulation observed intermediate state (crimson sticks) with L4 in loop-in conformation, compared to bound conformation (teal). (C) The differential residue contact probabilities with intermediate state compared to bound state. The color scale crimson-white-teal represents residue-GDM contacts in intermediate-only, common and bound-only respectively. (D) The implied timescales exhibiting three distinct ...

... states (blue, green, red). (E) The TICA based free energy re-weighted by MSM and its corresponding statemap. (F) The MSM binding pathways and their probabilities. (G) The three-state binding pathway identified by MSM. (H) The MSM derived binding kinetics. The k_{off} was estimated via infrequent metadynamics. (I) The characterization of MSM binding states via different metrics and τ -RAMD. (J) The binding mechanism observed in one of the replicates of funnel metadynamics, demonstrating $U \rightarrow I \rightarrow B$ pathway. The GDM molecule is represented via an arrow as demonstrated in inset. The final bound pose by funnel metadynamics was shown compared to crystal bound pose in inset. (K) The distance, Δ angle and RMSD of GDM during one of the funnel-metadynamics simulations. The arrowhead highlights the various intermediate states achieved during unsuccessful binding attempts. (L) The free energy landscape of GDM binding to N-Hsp90 via funnel metadynamics. The shaded width of curve represents the error bar measured over all replicates exhibiting successful binding.

Results and Discussion

Inhibitor binding to N-Hsp90 via an intermediate state

We began by investigating the binding mechanism of the inhibitor Geldanamycin (GDM) to N-Hsp90 through unbiased binding simulations. These simulations are widely recognized for their ability to recover binding mechanisms with atomistic detail, allowing ligand molecules in the unbound state to search for binding modes and ultimately bind to the native pocket as observed in crystal snapshots.^{26,33,34} In this study, the simulation box was filled with a single copy of N-Hsp90 (residues 9-236, Fig. S1), randomly surrounded by four GDM molecules (corresponding to a 10 mM concentration), positioned away from the Hsp90 target in the unbound state. Eight independent replicates were performed, each ranging from 0.5 to 6 μ s in duration (methods). The RMSD of GDM molecules with respect to their bound conformations was used as a metric to monitor the binding process.

Despite our efforts, the GDM molecules did not attain the crystal-bound pose (PDB-1YET) in any of the binding simulations, which is not entirely surprising given the slow binding nature of the large GDM inhibitor, a timescale potentially beyond the reach of conventional MD simulations. However, in two of the eight unbiased binding simulations, a stable binding mode near the native pocket was observed (Fig. 2A). We confirmed that this pose was not transient or unstable, as it persisted over multiple microseconds of simulation. A closer examination entailed that the GDM molecules encountered the N-Hsp90 in loop-in conformation of L4, whereby its further path to the bound state is blocked by L4 residues like L107, A108 and T109 (Fig. 2B). Although the crystal bound state and the simulation observed intermediate exhibit minor positional differences, the two states are significantly different in terms of residue interactions (Fig. 2C). A comparison of contact probabilities demonstrate that the bound state is characterized by specific ligand-receptor contacts buried within the basal beta sheets of the binding pocket and helices $\alpha 6$ - $\alpha 7$, while the intermediate state shows increased surface contacts, particularly with L3, $\alpha 4$ and L4. Due to its proximity to the binding pocket, this observed binding mode may serve as a potential intermediate state or an encounter-complex in the binding mechanism. This observation aligns with previous biochemical assays that indicated a two-step binding mechanism¹⁶ involving a potential intermediate state.

Although a long-lived intermediate state was consistently observed in multi-microsecond unbiased molecular dynamics (MD) simulations, transitions from this intermediate to the fully bound state were not captured within the simulation timescales. Upon extending the trajectories, this intermediate either dissociated back to the unbound state or persisted throughout the simulation, indicating a kinetic barrier. To quantitatively resolve the substrate-binding mechanism, and in particular to characterize the transitions between key intermediates along the binding pathway, we constructed a Markov state model (MSM).³⁵

To this end, the initial binding trajectories (Fig. 2A) were augmented with a large set of adaptively sampled MD simulations. Specifically, we initiated short unbiased simulations from multiple starting configurations, including the unbound state, the putative intermediate identified in earlier trajectories, and the crystal structure of the bound state (PDB ID: 1YET; Methods, SI). To further supplement the sampling of rare transitions, we employed multiple rounds of enhanced sampling using τ -RAMD^{36,37} and multiple rounds of adaptive targeted sampling (methods) to

generate 82 seed conformations in sparsely populated space. These were subjected to short-time scale unbiased MD simulations scheme. In total, 108 simulation trajectories were generated (Table S1,S2), yielding an aggregated unbiased sampling time of 38.5 μ s, which formed the basis for MSM construction.

The MSM was built using four key features derived from analysis of simulation data: (i) the minimum heavy atom distance between N-Hsp90 and the GDM molecule, (ii) the center-of-mass distance of GDM from the binding pocket (Fig. S1B), (iii) the RMSD of GDM relative to the bound pose, and (iv) the orientation of GDM with respect to the crystallographic binding conformation (Fig. S1 C). Kinetic decomposition revealed the presence of three dominant states, i.e., unbound (U), intermediate (I), and bound (B), with transitions occurring on well separated timescales, consistent across varying MSM lag times (Fig. 2D-H, S2). The primary binding pathway followed a two-step mechanism: $U \rightarrow I \rightarrow B$, with a pathway probability of 0.88 ± 0.01 , in agreement with prior experimental reports suggesting a two-step binding process for GDM binding to N-Hsp90.¹⁶ A minor direct binding pathway ($U \rightarrow B$) was also detected, but it accounted for only 0.12 ± 0.01 of the total pathway probability. (Fig. 2F) The intermediate state (I) acted as a key kinetic bottleneck, with a committor value of 0.12 ± 0.03 , (committor of U=0 and B=1) indicating frequent reversion to the unbound state, as observed in one unbiased trajectory (Fig. 2A).

From the MSM, the association rate constant (k_{on}) was estimated to be $(4.61 \pm 0.38) \times 10^6 \text{ M}^{-1} \text{ s}^{-1}$, consistent with typical small molecule-protein binding rate constant.³⁸ Notably, the transition from the intermediate to bound state ($I \rightarrow B$) was identified as the rate-limiting step, with a characteristic timescale of $18.34 \pm 1.92 \mu$ s, significantly longer than the $3.54 \pm 0.36 \mu$ s required for the $U \rightarrow I$ transition. In hindsight, this large disparity in timescales rationalizes the absence of observed $I \rightarrow B$ transitions in the initial 4-5 μ s trajectories (Fig. 2A), which were seeded from the unbound configuration.

Despite extensive adaptive sampling, the bound state remained under-sampled in unbiased simulations. We surmised that this is due to the inference of its exceptionally long residence time, on the order of seconds, rendering brute-force MD infeasible for quantifying the off-rate constant. Accordingly, we initiated a large replicate of infrequent metadynamics simulations³⁹ from the bound pose (methods), an enhanced sampling strategy well-suited for rare-event kinetics, especially known

for estimating substrate residence time and the off-rate constant (k_{off}). Analysis of these trajectories yielded a k_{off} of $(8.74 \pm 1.02) \times 10^{-2} \text{ s}^{-1}$ (Fig. 2H, S3), corresponding to a predicted residence time of $11.44 \pm 1.34 \text{ s}$. Combining the MSM-derived k_{on} and the metadynamics-derived k_{off} , we estimate a dissociation constant (K_d) of $18.97 \pm 2.71 \text{ nM}$, in close agreement with the experimentally reported value of approximately 9.0 nM .¹⁶

To further dissect the molecular characteristics and kinetic roles of the MSM-derived states, unbound (U), intermediate (I), and native bound (B), we performed additional structural and energetic analyses in conjunction with unbiased simulation data (Fig. 2G,I). For each state, we quantified three key descriptors: (i) the distance of the GDM ligand from the binding pocket, (ii) its RMSD relative to the crystallographic pose, and (iii) the orientation deviation (ΔAngle) compared to the native bound conformation. Additionally, we evaluated the relative kinetic stabilities of these states using τ -RAMD simulations^{36,37} (Fig. S4,S5).

In the unbound state (U), the GDM molecule remains distant from the binding site and explores a broad range of orientations and conformations, as expected for a freely diffusing ligand. As GDM approaches the binding pocket, it docks in multiple orientations, many of which are non-productive for progression to the bound state. This initial docking forms a metastable encounter complex, corresponding to the intermediate state (I), which acts as a kinetic bottleneck along the binding pathway. Importantly, the transition from I to B is further gated by the conformational state of the loop region surrounding the binding pocket. The loop-in conformation of N-Hsp90 further constrains GDM’s trajectory toward the bound state. Our simulations suggest that GDM frequently approaches the pocket in varied orientations but lingers in the intermediate state until achieving a selective orientation conducive to transitioning to the bound state (B), characterized by a “loop-out” conformation. Across independent binding events, the ligand was observed to engage the binding pocket from diverse angles but frequently remained in the intermediate state until a favorable orientation was achieved to permit transition to the fully bound conformation (Fig. 2G,I). An independent analysis of the relative residence times of the intermediate (I) and bound (B) states, using the unbound (U) state as a baseline, as determined from τ -RAMD simulations, confirmed their distinct kinetic stabilities. The intermediate state is a relative less stable but long lived encounter complex, formed by GDM before transitioning to the stable bound state.

Finally, we independently validated the GDM binding mechanism to N-Hsp90 using an alternative enhanced sampling approach funnel metadynamics⁴⁰ (Fig. 2J-L). This method applies a funnel-shaped restraining potential that restricts sampling away from the binding pocket while allowing full configurational freedom near the binding site, thereby accelerating the exploration of productive binding events. In these simulations, we used the distance between the GDM molecule and the binding pocket as the collective variable and initiated binding from the loop-out conformation of the L4 region of N-Hsp90 (Fig. S6). In contrast to the unbiased MD simulations, funnel metadynamics successfully captured transitions to the bound state, achieving final ligand poses within 1 Å RMSD of the crystallographic structure (PDB: 1YET).²⁸ Across eight independent replicates, five successful binding events were observed, yielding a binding free energy of $\Delta G = -10.74 \pm 1.07$ kcal/mol. This value is in close agreement with the experimental binding free energy of -10.97 kcal/mol corresponding to the reported dissociation constant ($K_D = 9$ nM).¹⁶

The binding pathway reconstructed from these simulations supported a three-state mechanism, akin to the MSM derived kinetic model described earlier (Fig. 2J-L, S6). The GDM molecule initially approached the binding pocket in diverse orientations, many of which were geometrically incompatible with productive binding. As a result, the ligand frequently became transiently trapped in a metastable intermediate state, leading to multiple unsuccessful binding attempts (Fig. 2K). Notably, orientation was not used as a biasing variable in these enhanced sampling simulations, indicating that the emergence of the intermediate state was an intrinsic feature of the binding landscape. Productive binding occurred only when the ligand attained a favorable orientation that enabled it to transition into the bound conformation.

To further probe the role of protein conformational gating, we repeated the funnel metadynamics simulations using the loop-in conformation of the L4 region. In these simulations, consistent with results from unbiased trajectories, the GDM molecule approached the binding pocket and formed intermediate states but was unable to complete the transition to the bound state (Fig. S7). Across eight independent replicates, no successful binding events were observed, despite proximity to the pocket, the ligand ultimately dissociated in each case (Fig. S7). These findings reinforce that both the ligand’s orientation and the protein’s conformational state, particularly the gating by the L4 loop, are essential prerequisites for successful binding.

In addition to the loop-in and loop-out configurations, N-Hsp90 is also known to adopt a helical conformation of L4, typically associated with specific inhibitor bound complexes but not observed with GDM or its analogs (see next section).²⁸ More recently, an ‘excited’ state of N-Hsp90 has been captured in both apo and inhibitor-bound forms^{20,21} (Fig. 1). Unlike the aforementioned ground states, this excited conformation is thought to be a post-binding structural transition, as seen in full-length Hsp90 complexes.⁴¹ To test whether this excited state plays any role in ligand recognition, we simulated binding and unbinding events using structural models from PDB ids 8B7J and 7UR3 (Supplementary results-I, Fig. S8). As expected, no binding or unbinding was observed in this conformation, the binding pocket is occluded by helices $\alpha 5$ and $\alpha 6$, precisely in the region where the intermediate state typically forms. These results indicate that, though the excited state surely serve a regulatory and dimerizing function following ligand engagement and may present within apo state, it remains a matter of research^{21,41–44} and it is unlikely to participate in the initial binding, at least for GDM.

Inhibitor Binding to N-Hsp90 Alters the Conformational Landscape of the Loop-4 Region, Supporting Both Conformational-selection and induced-fit

After establishing the two-stage substrate binding mechanism in Hsp90, we next focused on characterizing the key changes in the protein’s conformational space upon inhibitor binding to the native pocket, relative to the apo form. Adjacent to the native binding pocket of Hsp90, the loop-4 (L4, between $\alpha 4$ and $\alpha 5$) region is known to exhibit a remarkably high degree of conformational heterogeneity^{22,27} as broadly depicted in Fig. 1, spanning loop-in, loop-out, loop, helical, and excited conformations. Notably, multiple sub-conformations of Hsp90 have been experimentally resolved within the broader categories shown in Fig. 1. These conformations are highly sensitive to external factors, including the presence and type of inhibitor. For instance, GDM is known to bind via a loop-out conformation. In general, the L4 region of Hsp90 exhibits a highly dynamic conformational landscape that is strongly modulated by ligand binding; and not all conformations are simultaneously accessible under a given condition. To elucidate the conformational responses

to inhibitor binding, we performed extensive unbiased molecular dynamics simulations of both apo ($6 \times 4 \mu s$) and GDM-bound ($4 \times 4 \mu s$) states.

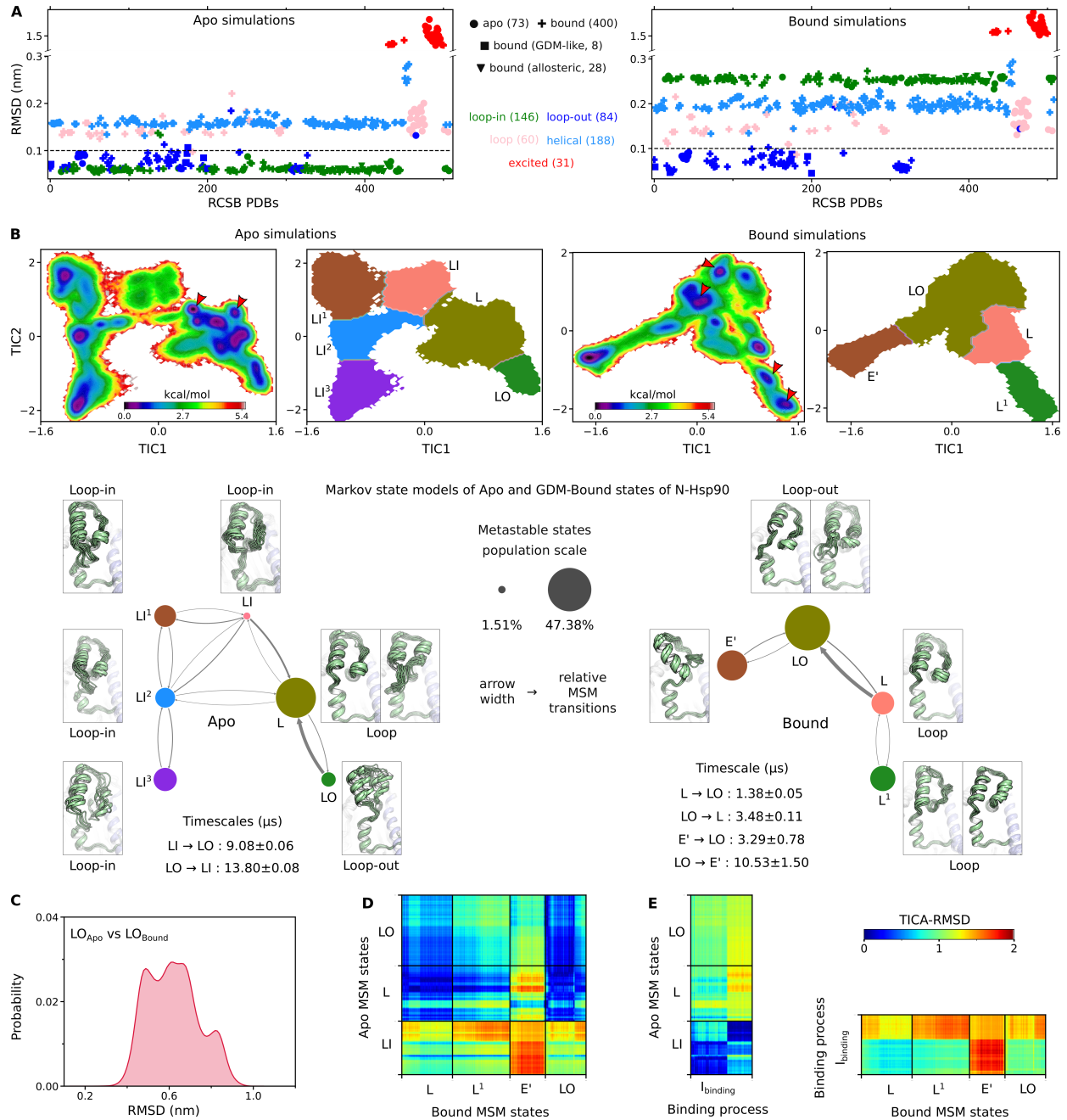


Figure 3: Conformational shift in N-Hsp90 by inhibitor binding: (A) The closest RMSD of $\alpha 4 + L4 + \alpha 5$ observed in apo and bound simulations with respect to all the PDB structures. (B) The markov state models of apo and bound simulations. (C) The probability distribution of all one-to-one RMSD between loop-out conformations apo and bound MSM. (D,E) The TICA-based RMSD between the protein conformations of apo MSM, bound MSM and intermediate state of binding mechanism. Note that only small random subset of conformations are used in for this comparison, due to memory issues. The combined TICA was performed on all datasets, apo+bound+binding+adaptive simulations for this comparison.

We first characterized the simulated conformational ensembles by comparing them against the

experimentally reported structural diversity. A total of 422 crystal and NMR structures have been reported for Hsp90 in the RCSB database. After filtering for missing residues, 509 complete protein chains were retained for analysis (methods). We evaluated whether these experimentally observed conformations are captured in our simulations (Fig. 3A). Notably, the loop-in (146 chains) and loop-out (84 chains) conformations were sampled within 1 Å RMSD in the apo simulations, encompassing both apo and inhibitor-bound structures. Perturbation response scanning of the apo trajectories revealed that perturbing most of the residues can induce *loop-in* $\rightarrow$ *loop-out* transitions and vice versa, suggesting a propensity for spontaneous interconversion (Fig. S9). *Loop* conformations associated with non-GDM ligands were similarly recapitulated within 1.5 Å RMSD. Interestingly, the *loop* conformation in the apo state of the R60A mutant (PDB: 8B7I) was sampled around 2 Å RMSD—slightly higher than other loop conformations. To further validate our simulations, we also compared simulated structures to the associated NMR NOE data²¹ (methods, Table S3) by quantifying average violations. Most frames from apo trajectories exhibited average NOE violations below 2 Å (Fig. S10), although some frames with larger violations suggest that our simulations also explore conformations beyond the dominant loop states. Furthermore, though the entropically restricted helical conformations sample within 2 Å RMSD in our simulations, the perfect helicity was not observed. Using FAST adaptive sampling⁴⁵ approaches (Methods), we concluded that the perfect helical conformation is not accessible in apo state, potentially due to its entropic penalty (Supplementary Results-II, Fig. S11). This is consistent with the fact that helical conformations are experimentally observed only in the presence of inhibitors that stabilize L4, and not in apo state. Additionally, the recently reported excited state conformations were not sampled, due to their extremely large timescales of nearly 12 ms, which are not accessible by unbiased MD. Overall, most of the experimentally determined conformational heterogeneity on N-Hsp90 is sampled within the simulations at < 2 Å.

GDM-bound simulations also revealed a rich conformational ensemble but with a notable absence of the loop-in conformations (Fig. 3A), consistent with available crystal structures. These results confirm that the conformational landscape of Hsp90 is highly sensitive to ligand binding and emphasize the importance of long timescale simulations for capturing these structural transitions. To characterize the underlying kinetics and dominant states, we constructed Markov state models

(MSMs) for both apo and GDM-bound ensembles, with a particular focus on the L4 dynamics (methods; Fig. 3B).

The MSM derived state space of apo Hsp90 resolved three principal conformational categories, loop-in (LI, LI^1, LI^2, LI^3), loop (L), and loop-out (LO), comprising six metastable states in total (Fig. 3B, S12). In contrast, the GDM-bound MSM state space lacked any loop-in conformations and instead featured four metastable states: loop (L, L^1), loop-out (LO), and a partially populated state reminiscent of the recently proposed excited state (E') (Fig. 3C). The $\alpha 5$ and $\alpha 6$ helices were also highly dynamic in simulations, such that $\alpha 5$ sampled different orientations while small $\alpha 6$ sample folded-unfolded conformations (Fig. 3B). These are the signatures of the recently proposed excited state, whereby $\alpha 6$ is unfolded and $\alpha 5$ is significantly shifted from its location (PDB ids 8B7J, 7UR3).^{20,21} All the L4 conformations, observed for apo and bound MSM states, possessed further sub-states based on $\alpha 5$ - $\alpha 6$ conformations, either as separate metastable states or multiple minimas within a free energy space (Fig. 3B).

The comparison between the apo and GDM-bound conformational ensembles of Hsp90 offers mechanistic insights into whether GDM binding proceeds via conformational-selection, or induced-fit mechanism. The L4 conformational transitions within the apo state occurred at $9.08 \pm 0.06 \mu s$ ($LI \rightarrow LO$) and $13.08 \pm 0.08 \mu s$ ($LO \rightarrow LI$) timescales (Fig. 3B), which are smaller than binding timescales ($21.7 \pm 1.78 \mu s$, Fig. 2H). The loop conformations of bound state is common to their counterparts in apo state. Furthermore, the dominant loop-out conformation ($49.34 \pm 0.40\%$) of bound state pre-exists in apo state as a minor conformation ($6.4 \pm 0.02\%$). These observations strongly hinted at the conformational-selection binding mechanism.

However, the dominant loop-out conformation in bound state is not similar to its counter-part in apo state MSM, deviating from conformational-selection mechanism. A quantitative comparison of root-mean-square deviations (RMSD) of L4 residues in LO_{Bound} and LO_{Apo} conformations indicate structural differences as large as 6 Å (Fig. 3C,D). Specifically, the L4 region in the LO_{Bound} state adopts a more extended loop-out conformation than its barely out LO_{Apo} counterpart, indicating that GDM binding may further remodel the pre-existing conformation. On the combined TICA space, the LO_{Bound} and LO_{Apo} conformations are not similar, and bound state also possesses E' conformation not accessible in apo state (Fig. 3E). Additionally, the intermediate binding state (I

state in Fig. 2) retains a loop-in L4 conformation that closely resembles the apo MSM states and not the bound MSM states (Fig. 3E), suggesting that loop-out conformations arise later along the binding pathway. These observations hinted at the induced-fit binding mechanism. Overall, the observations collectively point toward a mixed mechanism in which both conformational-selection and induced-fit contribute to the recognition and binding of inhibitor to Hsp90.

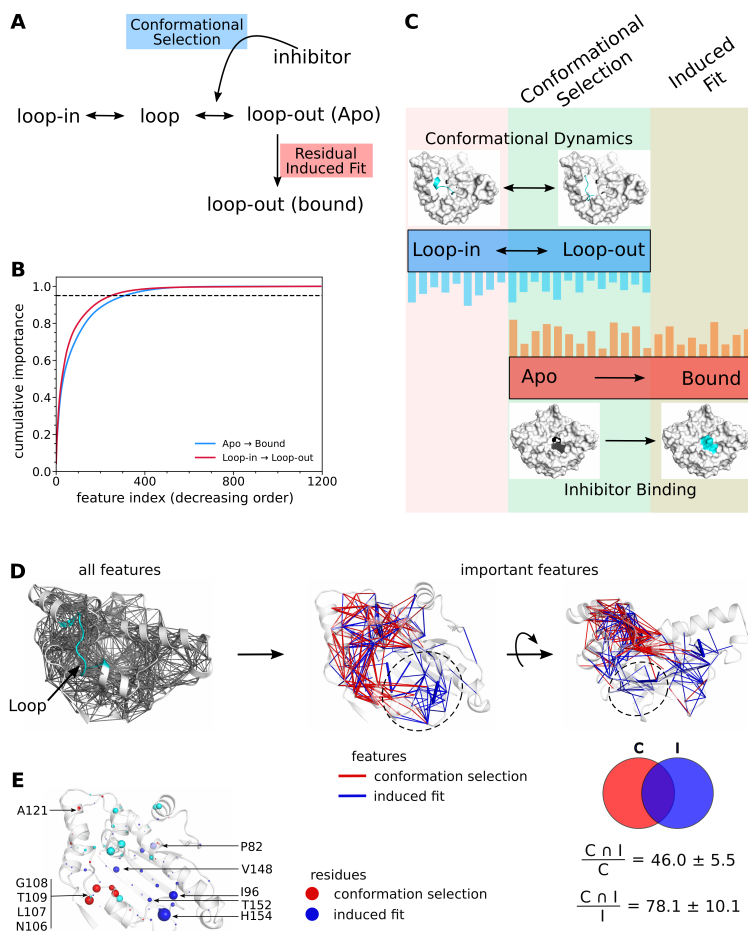


Figure 4: Conformation-selection vs induced-fit in N-Hsp90: (A) Residual induced-fit scheme for Hsp90. (B) Mean cumulative feature importance scores obtained by trained RF classifiers. (C) Schematic for segregation of conformational-selection and induced-fit. (D) All and selected features based on feature importance scores and colored according to conformational-selection (red) or induced-fit(blue). (E) Important residues involved in conformational-selection and induced-fit mechanisms in N-Hsp90.

A Machine Learning Classifier Quantifies Feature Importance and the Extent of Conformational-selection and induced-fit in the Substrate Recognition Process

The preceding section indicates that GDM binding alters the conformational space of N-Hsp90 in a complex manner, suggesting potential involvement of both conformational-selection and induced-fit mechanisms (Fig. 3). This dual mechanism is not uncommon in biophysics; various systems, including proteins^{46,47} and RNA,⁴⁸ have been shown to exhibit diverse combinations of conformational-selection and induced-fit mechanisms. For inhibitor binding to Hsp90, we propose a similar scenario, where GDM binding initially follows a conformational-selection pathway by selectively binding to the *loop* $\leftrightarrow$ *loop-out* conformation of apo N-Hsp90 (Fig. 2, S6,S7). This binding is subsequently followed by conformational changes induced in the loop-out state, resulting in a different loop-out conformation (Fig. 4A). This phenomenon is referred to as “residual induced-fit” in the literature.⁴⁹ To explore this, we developed a machine learning classifier, specifically a Random Forest (RF) model, to exclusively detect the conformational changes associated with GDM binding through conformational-selection and induced-fit mechanisms.

In addition to learning labels, RF models can weigh input features based on their ability to classify the labels. These weights, known as ‘feature importance scores,’ have been used to identify features involved in protein allostery, residues differentiating MSM states and others insights from simulations.^{32,50–52} Similarly, we employed RF-based feature importance scores to detect significant conformational changes in N-Hsp90. To define the global conformation of N-Hsp90, native contacts (minimum distances in simulations) between residue pairs within 10 Å in the crystal structure were used. These contacts capture conformational changes across the entire protein, not just in the loop-4 region (as in MSM). Thus, each simulated conformation was represented by numerical values, which served as input features for RF training, with each feature representing a conformational change between two residues in terms of minimum distance (methods).

In the first part of our analysis, input features from apo simulation trajectories were used, and each conformation was labeled with MSM-derived states: loop-out, loop-in, and loop, as shown in Fig. 3C. The RF models were successfully trained, achieving an accuracy of 0.99 ± 0.00 ($10e-7$)

and an F1 score of 0.98 ± 0.00 ($10e-4$) (Fig. S13). Feature importance scores revealed that out of 1194 features, only 245 ± 2 features corresponded to 95% of the conformational dynamics defining the MSM states (labels). Replicating the RF training on 50 cross-validated samples indicated that 206 of these 245 ± 2 features consistently emerged as important across all samples (Fig. 4B, S13). These 206 features represent the set of conformational changes that define the intrinsic dynamics of the apo states, referred to as ‘conf1,’ with their associated feature importance scores reflecting their relative contributions.

In the second part, another RF classifier was trained on input features from both apo and bound simulations. The input features were labeled as apo or bound (0 and 1), based on the simulations. The training was successful, with an accuracy of 0.99 ± 0.00 and an F1 score of 0.99 ± 0.00 (Fig. S13). According to the feature importance scores, 304 ± 3 features out of 1194 accounted for 95% of the conformational difference between apo and bound states. Of these, 256 features were consistent across all 50 cross-validated replicates (Fig. 4B). These features represent the set of conformational changes that define the comparative dynamics between apo and bound states, i.e., the GDM binding process, referred to as ‘conf2.’

We surmized that the conformational changes associated with GDM binding (‘conf2’) that are also part of the intrinsic apo dynamics (‘conf1’) represent the features responsible for conformational-selection, as these changes can spontaneously occur within the apo state (Fig. 4C). Conversely, the unique conformational changes in ‘conf2’ that are not observed in the apo state (‘conf1’) likely represent the features responsible for the induced-fit mechanism. The remaining conformational changes in ‘conf1’ that are not part of ‘conf2’ would not have any role in the GDM binding process. This allowed us to spatially locate and quantify the different conformational changes. Superimposing the conformational changes of ‘conf1’ and ‘conf2’ on the N-Hsp90 structure revealed an almost segregated presence of conformational-selection (common in ‘conf1’ and ‘conf2’, shown in red) and the induced-fit mechanism (specific to ‘conf2’, shown in blue) (Fig. 4D). The features responsible for conformational-selection are specifically located in the L4 region, $\alpha 5$, $\alpha 6$, and $\alpha 7$, while the features responsible for induced-fit are mostly located in the L3 and buried binding pocket regions (Fig. 4D). Summing the feature importance scores, $46.0 \pm 5.5\%$ of the conformational changes in ‘conf1’ are common with ‘conf2,’ representing conformational-selection, which accounts for $78.1 \pm 10.1\%$ of

‘conf2.’ This suggests that the remaining 28% of ‘conf2’ features correspond to the induced-fit mechanism. As a further check, we repeated the analysis using input features only from loop-4 (as in MSM of Fig. 3) and observed similar quantifications: common conformational changes of $42.9 \pm 6.8\%$ of ‘conf1’ represents $67.9 \pm 12.3\%$ of ‘conf2’ conformational changes.

To make this analysis more meaningful at the residue level, we dissected the conformational changes further. Each line in Fig. 4D represents a feature, i.e., the minimum distance between two residues. The feature importance score of each feature was equally divided between its two residues, and these scores were summed to represent each residue’s relative importance. We observed three types of residues: (a) those associated only with the conformational-selection route (red), (b) those associated only with induced-fit (blue), and (c) those associated with both (cyan) (figure 4E). The residues predicted to be involved in conformational-selection are exclusively located in the L4 region (N106, L107, G108, T109), which is also the most extensively investigated region of N-Hsp90. In particular, L107 has recently been identified as the most important residue for its entropic contribution.²⁷ This is consistent with previous reports that the L4 region exhibits conformational heterogeneity within the apo state, with inhibitors specifically binding to some of these conformations.²⁷ The residues predicted to be involved in the induced-fit mechanism are primarily located in the L3 region and within the binding pocket. Coincidentally, this L3 region is where a binding intermediate was observed (Fig. 1), validating the existence of an intermediate state. This observation aligns with previous reports suggesting the existence of an intermediate state and an induced-fit pathway,¹⁶ based on the inference that the induced-fit mechanism proceeds through intermediate steps. Notably, the simulations containing the intermediate state were not used in the above analysis, yet induced-fit in the L3 region was independently observed. The cyan residues in $\alpha 6-7$ are involved in both conformational selection and induced-fit mechanisms. This is consistent with Fig. 3 that conformational heterogeneity of this region is observed in both apo and bound states, with but with some differences like metastable state E’.

Conclusion

This study provides an inside-look at the complexity inherent in inhibitor recognition process of N-Hsp90, entangled with its own conformational plasticity and how machine learning can augment the dissection of full process. Using a combination of unbiased molecular dynamics, enhanced sampling, and adaptive simulation strategies, we reconstruct a detailed binding mechanism encompassing unbound, intermediate, and bound states. This mechanistic picture is supported by Markov state models (MSMs), unbinding kinetics via infrequent metadynamics, and residence time estimates from τ -RAMD. A specific proposal of this work is the identification of a putative intermediate state. This intermediate state was observed through unbiased binding simulations, which were conducted without any prior bias towards the binding site. These simulations are well-regarded for their ability to recover binding paths accurately.^{26,33,34} Together, these insights establish a rigorous computational framework for probing highly dynamic chaperones and provide guiding insights for the rational design of next-generation Hsp90 inhibitors targeting conformational sub-states.

The generalizability of our findings rests on the assumption that other known and emerging inhibitors are likely to follow binding mechanisms similar to that elucidated for GDM. Given the structural diversity of GDM derivatives and the vast number of potential inhibitor candidates, it is impractical to investigate each compound individually. To this end, we selected GDM’s parent scaffold for many ATP-competitive inhibitors currently under development—as a representative system.^{12–15} Since these inhibitors commonly target the conserved ATP-binding pocket, we anticipate that their binding pathways will largely recapitulate the mechanistic features observed for GDM.

While individual inhibitors may differ in the precise nature of their interactions in the bound or intermediate states, our results suggest a broadly applicable three-state binding pathway, in which the intermediate conformation serves as a critical kinetic and structural bottleneck. This intermediate-centric framework may thus offer a transferable mechanistic basis for guiding inhibitor design across a wide range of Hsp90-targeting compounds.

From a biophysical perspective, the observed mechanism challenges the conventional binary view of protein-ligand binding pathways as either conformational-selection or induced-fit. Our

results reveal that inhibitor binding induces significant changes in the conformational space of Hsp90, which can be attributed to the simultaneous presence of both conformational selection and induced-fit mechanisms. This study provides a quantitative estimate of the extents to which these mechanisms contribute to inhibitor binding. The coexistence of these two binding regimes is not uncommon,^{46–49} yet it remains challenging to estimate their relative contributions accurately.⁵³ To the best of our knowledge, no existing computational methods are available to achieve this level of detail. The residual induced-fit mechanism elucidated here is significant, as it highlights the dynamic interplay between intrinsic protein flexibility and ligand-induced stabilization.

From a mechanistic standpoint, different N-Hsp90 inhibitors are likely to vary in the relative contributions of conformational selection and induced fit. However, our results suggest that the broader assignment of structural elements to these mechanisms may be generalizable across most, if not all, N-Hsp90 inhibitors. Specifically, loop L4 predominantly participates in conformational-selection, while loop L3 and the basal β -sheet region are primarily engaged through induced-fit, and the $\alpha 6$ - $\alpha 7$ segment contributes to both mechanisms. This classification aligns with prior findings from a study of 20 N-Hsp90 inhibitors, which identified L4 as a key entropic driver of binding, with residue L107 as a principal contributor.²⁷ Such agreement reinforces the general applicability of our conclusions. Nevertheless, quantitatively distinguishing simultaneous contributions from both mechanisms remains experimentally challenging and, to our knowledge, has not yet been accomplished for Hsp90. Our framework also allows for the prediction of binding mechanism profiles across inhibitor classes. Inhibitors that target the loop-out conformation, similar to GDM (a loop-out binder), likely invoke both conformational-selection and induced-fit. In contrast, inhibitors that bind the helical conformation are expected to exhibit a predominantly induced-fit mechanism, as this conformation is inaccessible in the apo ensemble and likely stabilized only upon inhibitor engagement. Meanwhile, inhibitors that bind to pre-existing loop-in or loop conformations are expected to rely primarily on conformational-selection, with only minor contributions from induced-fit in the basal β -sheet region. Within each class, individual inhibitors may differ in the relative balance between these two binding paradigms, depending on their chemical and structural properties.

Beyond structural insights, our detailed characterization of the intermediate state and its unique

residue-level interactions highlights its potential as a novel target for drug design. Despite significant effort, many Hsp90 inhibitors, including GDM, have failed to meet the pharmacological thresholds for cytotoxicity and bioavailability. Targeting the intermediate state offers an alternative strategy: molecules designed to stabilize this state could enhance association kinetics by lowering energy barriers along the binding pathway. Conversely, inhibitors that selectively destabilize the intermediate could prolong residence time by impeding the dissociation process, a growing focus in structure-based drug design.^{27,54} By integrating free energy landscapes, conformational state decomposition, and feature attribution analyses, our study delineates the roles of specific residues and structural motifs in substrate recognition and binding. This framework not only informs the rational design of new Hsp90 inhibitors but can be generalized to other biomolecular systems exhibiting conformational heterogeneity and multi-step binding, such as intrinsically disordered proteins or membrane-associated receptors. Overall, this work supports an emerging paradigm in molecular recognition: that productive ligand binding does not require rigid conformational stabilization, but rather, precise modulation of an accessible conformational ensemble across multiple metastable states.

Methods

Simulation Methods

All atom molecular dynamics simulations were performed for N-Hsp90 (residues 9-236) via GROMACS-version20xx (20.5,20.1,24.3),⁵⁵ patched with plumed.⁵⁶ The simulations were designed to study inhibitor binding mechanism to N-Hsp90, and to sample the its conformational heterogeneity. The N-Hsp90 was parameterized with charmm36⁵⁷ forcefield, GAAMP⁵⁸ based parameters were generated for inhibitor GDM, and TIP3P⁵⁹ model for water molecules. The simulations were performed at 310.15 K temperature and 1 atm pressure. All simulation and analysis methodology is detailed in supplementary information. Briefly, nearly 1000 replicates accounting for more than 100 μs simulations were performed, categorized as below:

- **Unbiased binding simulations** Eight replicates of binding simulations^{26,33} of GDM to N-Hsp90 were performed whereby, the four molecules of GDM starts from unbound state and unbiasedly search for their binding pocket (Table S1).
- **State specific simulations:** The N-Hsp90 was simulated in apo, intermediate and GDM bound states (Table S1).
- **τ -RAMD simulations:** The relative residence times of intermediate and bound states were measured by stochastic expulsion forces applied to GDM molecules bound at the desired state, through τ -RAMD simulations,^{36,37} via its gromacs implementation.⁶⁰
- **Targeted simulations:** High temperature (500 K) simulations were performed to generate the starting conformations for rare states or transitions, i.e., bound to intermediates, ground to excited states. Suitable restraints were applied to prevent protein misfolding.
- **Adaptive sampling:** Short 100 ns simulations were performed from the 50+32 starting configurations generated by τ -RAMD and targeted simulations, to supplement the rare transitions during the binding process.
- **Funnel metadynamics simulations:**^{40,61} Binding pathways and free energy profiles were computed using a cone-cylinder funnel restraint setup for loop-in, loop-out, and excited conformations. Bias potentials were applied along ligand-pocket distances.
- **Infrequent metadynamics simulations:**^{39,62} The off-rate constant of GDM was estimated via infrequent metadynamics in ground state:loop-out and excited state conformations.
- **FAST adaptive sampling simulations:**⁴⁵ Bias free adaptive sampling was used to model transitions among L4 loop conformations (loop-in, loop-out, and helical) based on a structural scoring metric. Additional GDM bound excited state was modelled using these simulations.

Analysis Methods

A large set of post-simulation analysis were conducted to uncover binding mechanisms, conformational transitions, and structural determinants. All the methods are detailed in supplementary

information. Briefly, we constructed or trained:

- **Markov State Models (MSMs):**^{35,63,64} Three MSMs were built to study the (i) GDM binding process (38.5 μ s), (ii, iii) loop dynamics in apo (23.3 μ s), and inhibitor bound (16 μ s) states (Table S2).
- **Machine learning classifiers:** Two random forest classifiers were trained each for native contacts, and L4-residue distance features. One model classified L4 conformational states within apo ensembles; the other distinguished apo from bound conformations. Feature importance was derived from 50 cross-validated models.
- **Perturbation response scanning (PRS):** Using linear response theory, PRS⁶⁵ quantified the sensitivity of L4 transitions to perturbations across the protein. Calculations were performed using the md-task suite on apo-state trajectories, with reference structures from PDB IDs 1YER and 1YES.
- **NOE distance violations:** Simulated distances between methyl-methyl proton centers were compared against 19 experimental NOE-derived constraints from the R60A mutant.²¹ Violations were averaged across apo trajectories and benchmarked against experimental bounds (Table S3).
- **Comparison with experimental structures:** A curated dataset of 509 chains from NMR and crystal structures (PDB entries of UniProt ID: P07900)⁶⁶ was used to evaluate the conformational space sampled in simulations. Structures lacking key residues or with anomalous geometries (e.g., PDBs 8JRA, 8JR6, 8JR7) were excluded.

Supplementary Information

Detailed simulation methods and models, analysis details including Markov state Models, Machine learning classifiers etc, Supplemental tables S1-S3 and figures S1-S13.

Data and code availability

The codes, input parameter files, GDM forcefield parameters, trained models used in this work are open sourced at <https://github.com/msahilgit/Hsp90> . All the raw MD simulation trajectories generated in this work shall be open-sourced on zenodo upon acceptance.

Acknowledgement

We acknowledge support of the Department of Atomic Energy, Government of India, under Project Identification No. RTI 4007. JM acknowledges Core Research grants provided by the Department of Science and Technology (DST) of India (CRG/2023/001426). We thank Dr. Sneha Menon and Subinoy Adhikari for discussions and their useful suggestions. This work was supported by shared computing resources obtained from TIFR Hyderabad, India.

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